

Connecting an AI Coding Agent to AWS

Proof of setup and verification: AWS Agent Toolkit with Claude Code

Environment: macOS · Claude Code (Opus 5) · Account: tanseer-personal · Region: us-east-1 (N. Virginia)

Overview

This document records the end-to-end process of connecting an AI coding agent (Claude Code) to a live AWS account using the AWS Agent Toolkit, and provides screenshot evidence that the connection works. Four checks were carried out: installation and configuration, basic connectivity, live (not memorised) data retrieval, and write permissions.

Installation and Configuration

The toolkit was configured from the terminal with a single command:

```
aws configure agent-toolkit
```

The installer then completed the following steps in about two minutes:

- Detected installed AI coding agents and found Claude Code at `~/claude/skills` (Cline, Codex, Cursor, Gemini CLI, Kiro, OpenClaw, OpenCode, Pi and Windsurf were not installed).
- Fetched and installed 23 default AWS skills, including `amazon-bedrock`, `aws-iam`, `aws-cdk`, `aws-serverless`, `aws-storage` and `aws-security`.
- Configured the AWS MCP server connection for Claude Code by updating `~/claude.json`.

```
aws configure agent-toolkit

Detecting installed AI coding agents...
✓ Claude Code - ~/.claude/skills
✗ Cline - ~/.cline/skills (not found)
✗ Codex - ~/.agents/skills (not found)
✗ Cursor - ~/.cursor/skills (not found)
✗ Gemini CLI - ~/.agents/skills (not found)
✗ Kiro - ~/.kiro/skills (not found)
✗ OpenClaw - ~/.openclaw/skills (not found)
✗ OpenCode - ~/.agents/skills (not found)
✗ Pi - ~/.pi/agent/skills (not found)
✗ Windsurf - ~/.agents/skills (not found)

Fetching default AWS skills...
Found: amazon-bedrock, aws-ai-ml, aws-auth, aws-billing-and-cost-management, aws-blocks, aws-cdk, aws-cloudformation, aws-compute, aws-containers, aws-database, aws-deployment, aws-iam, aws-messaging-and-streaming, aws-networking, aws-observability, aws-sdk-js-v3-usage, aws-sdk-python-usage, aws-sdk-swift-usage, aws-security, aws-serverless, aws-storage, launch-with-aws, signing-in-to-aws

Install 23 default AWS skills? [Y/n]: y

Installing 23 default AWS skills...
[1/23] amazon-bedrock
[2/23] aws-ai-ml
[3/23] aws-auth
[4/23] aws-billing-and-cost-management
[5/23] aws-blocks
[6/23] aws-cdk
[7/23] aws-cloudformation
[8/23] aws-compute
[9/23] aws-containers
[10/23] aws-database
[11/23] aws-deployment
```

Figure 1: Running aws configure agent-toolkit – agent detection and skill download

```
Installing 23 default AWS skills...
[1/23] amazon-bedrock
[2/23] aws-ai-ml
[3/23] aws-auth
[4/23] aws-billing-and-cost-management
[5/23] aws-blocks
[6/23] aws-cdk
[7/23] aws-cloudformation
[8/23] aws-compute
[9/23] aws-containers
[10/23] aws-database
[11/23] aws-deployment
[12/23] aws-iam
[13/23] aws-messaging-and-streaming
[14/23] aws-networking
[15/23] aws-observability
[16/23] aws-sdk-js-v3-usage
[17/23] aws-sdk-python-usage
[18/23] aws-sdk-swift-usage
[19/23] aws-security
[20/23] aws-serverless
[21/23] aws-storage
[22/23] launch-with-aws
[23/23] signing-in-to-aws

Skills installed to:
Claude Code - ~/.claude/skills

Configure AWS MCP server connection? [Y/n]: Y

AWS MCP server configured for:
✓ Claude Code - ~/.claude.json: updated

You can discover additional skills with 'aws agent-toolkit search-skills --search-query <text>'

2m 0s 10:24:20 AM
```

Figure 2: All 23 skills installed and the AWS MCP server configured for Claude Code

Basic Connectivity Test

To confirm that Claude Code could reach AWS through the MCP server, the agent was asked a simple question:

“What AWS Regions are available?”

The agent called the aws-mcp tool and returned all 37 AWS Regions, grouped by geography (US; Canada, Mexico and South America; Europe; Asia Pacific; Middle East, Africa and Israel), plus the partitioned GovCloud and European Sovereign Cloud

Regions. It also correctly noted that this is the full AWS catalogue, that opt-in Regions must be enabled per account, and offered to check which Regions are enabled on this account.

```

  /Users/tanseer

| Using Opus 5 (1M context) (from .claude/settings.json) · /model

What AWS Regions are available?

Called aws-mcp

• 37 AWS Regions:

US — us-east-1 N. Virginia · us-east-2 Ohio · us-west-1 N. California · us-west-2 Oregon

Canada / Mexico / South America — ca-central-1 Central · ca-west-1 Calgary · mx-central-1 Mexico Central · sa-east-1 São Paulo

Europe — eu-west-1 Ireland · eu-west-2 London · eu-west-3 Paris · eu-central-1 Frankfurt · eu-central-2 Zurich · eu-north-1 Stockholm · eu-south-1 Milan · eu-south-2 Spain

Asia Pacific — ap-northeast-1 Tokyo · ap-northeast-2 Seoul · ap-northeast-3 Osaka · ap-east-1 Hong Kong · ap-east-2 Taipei · ap-south-1 Mumbai · ap-south-2 Hyderabad · ap-southeast-1 Singapore · ap-southeast-2 Sydney · ap-southeast-3 Jakarta · ap-southeast-4 Melbourne · ap-southeast-5 Malaysia · ap-southeast-6 New Zealand · ap-southeast-7 Thailand

Middle East / Africa / Israel — me-south-1 Bahrain · me-central-1 UAE · il-central-1 Tel Aviv · af-south-1 Cape Town

Partitioned (separate credentials/endpoints) — us-gov-east-1, us-gov-west-1 (GovCloud) · eusc-de-east-1 (European Sovereign Cloud, Germany)

Note this is the full AWS catalog, not what your account has enabled — opt-in Regions (most of the newer ones: Hyderabad, Jakarta, Melbourne, Zurich, Spain, UAE, Tel Aviv, Cape Town, Hong Kong, Milan, Bahrain, Calgary, Mexico, etc.) need explicit enabling per account, and GovCloud/ESC are separate partitions. Want me to check which ones are actually enabled on your account?

* Cogitated for 17s · done 11:08 AM

Update available! Run: brew upgrade claude-code@latest

> Yes, check which ones are enabled on my account

>> auto mode on (shift+tab to cycle) · ← for agents

```

```

  /Users/tanseer

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What AWS Regions are available?

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Asia Pacific — ap-northeast-1 Tokyo · ap-northeast-2 Seoul · ap-northeast-3 Osaka · ap-east-1 Hong Kong · ap-east-2 Taipei · ap-south-1 Mumbai · ap-south-2 Hyderabad · ap-southeast-1 Singapore · ap-southeast-2 Sydney · ap-southeast-3 Jakarta · ap-southeast-4 Melbourne · ap-southeast-5 Malaysia · ap-southeast-6 New Zealand · ap-southeast-7 Thailand

Middle East / Africa / Israel — me-south-1 Bahrain · me-central-1 UAE · il-central-1 Tel Aviv · af-south-1 Cape Town

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Update available! Run: brew upgrade claude-code@latest

> █

>> auto mode on (shift+tab to cycle) · ← for agents

```

Figure 3: Regions query answered through aws-mcp (completed in 17 seconds)

Live Data Test: Reading Current Docs, Not Memory

The next step was to confirm that the agent answers from live AWS sources and the actual account, rather than from its training data. Two questions were used.

Latest EC2 instance families

“What’s the latest instance type family AWS added to EC2?”

The agent called `aws-mcp` five times and reported the Graviton5-based 9th-generation families: `R9g/R9gd` (generally available August 2026), following `C9g/C9gd` and `M9g/M9gd` (generally available June 2026). It confirmed that `c9g`, `c9gd`, `m9g`, `m9gd`, `r9g` and `r9gd` appear in `DescribeInstanceTypes` for `us-east-1` and `us-west-2`, and linked the source announcements. Because these releases are recent, the answer could only have come from current documentation.

```
Jakarta, Melbourne, Zurich, Spain, UAE, Tel Aviv, Cape Town, Hong Kong, Milan, Bahrain, Calgary, Mexico, etc.) need explicit enabling per account, and GovCloud/ESC are separate partitions. Want me to check which ones are actually enabled on your account?

* Cogitated for 17s · done 11:08 AM

What's the latest instance type family AWS added to EC2?

Called aws-mcp 5 times

• AWS Graviton5 – the 9th-generation *9g families. The most recent brand-new family to reach GA is R9g / R9gd (memory-optimized) in August 2026, following C9g / C9gd and M9g / M9gd, which both went GA in June 2026 (M9g had been in preview since December 2025).

What's new about them:
- Up to 25% better compute than Graviton4-based *8g; ~30% faster for databases, ~35% for web apps and ML
- 5x larger cache, fastest memory of any cloud instance per AWS
- 6th-gen Nitro System, first with the Nitro Isolation Engine
- d variants add local NVMe SSD

Availability: US East (N. Virginia, Ohio), US West (Oregon), Europe (Frankfurt, Ireland), Asia Pacific (Singapore, Sydney, Tokyo) – still expanding. I confirmed c9g, c9gd, m9g, m9gd, r9g, r9gd all appear in DescribeInstanceTypes for us-east-1 and us-west-2.

Worth noting alongside:
- P6-B300 (8x NVIDIA Blackwell Ultra, 2.1 TB GPU memory, 6.4 Tbps EFA) – newest accelerated family; Aug-Sep 2026 news is region expansion, not initial GA
- G7 / G7e (Blackwell RTX PRO) – GA Jan and June 2026
- C8line / M8line – network-optimized Intel, GA April 2026

Sources: C9g/C9gd, M9g/M9gd, R9g/R9gd

* Cogitated for 1m 3s · done 11:10 AM

Update available! Run: brew upgrade claude-code@latest

> Check if c9g is available in my account

>> auto mode on (shift+tab to cycle) · ← for agents
```

Figure 4: EC2 instance-family query answered from current AWS sources, with linked citations

Account-specific Bedrock quotas

“What are the current quota limits for Bedrock?”

The agent loaded the `amazon-bedrock` skill and pulled the applied quotas from the account itself rather than quoting defaults. In `us-east-1` it enumerated all 1,069 Bedrock quotas and compared the applied values against the AWS defaults. Key findings:

- Account-level ceiling: Cross-Model Account-Level Tokens Per Day is 150M (default 700M).
- Claude model limits are well below defaults, e.g. Sonnet 4.6 at 10 RPM and 6M TPM (default RPM is 10,000), making requests per minute the binding constraint.

- Several newer models (including Opus 5 and Sonnet 5) and all bedrock-mantle endpoint quotas are at zero until raised.
- Recommended actions: request quota increases, set maxTokens explicitly on every call, and confirm billing is in good standing.

These figures are specific to this account, which shows the agent is reading live account data rather than relying on memory.

```
What are the current quota limits for bedrock

• I'll pull the actual applied quotas from your account rather than quoting defaults.
• Skill(amazon-bedrock)
  ↳ Successfully loaded skill
  Called aws-mcp 2 times
• Bedrock exposes ~1,100 quotas per region, so let me narrow to one region at a time.
  Called aws-mcp

----- 1 new message -----

• Checked us-east-1 (Bedrock has 1,069 quotas per region, fully enumerated – applied vs. AWS default).

Headline: your account is quota-restricted across the board

Every quota that differs from the AWS default is lower, not higher – nothing in this account has been raised. This is the signature of a new or unverified account rather than a tuned one.

Account-level ceiling: Cross-Model Account-Level Tokens Per Day = 150M (default 700M) – this caps everything regardless of per-model limits.

Claude models you can actually use (us-east-1)

| Model      | RPM | TPM | Notes                               |
|------------|-----|-----|-------------------------------------|
| Sonnet 4.6 | 10  | 6M  | default RPM 1 new message (click) ⓘ |


Update available! Run: brew upgrade claude-code@latest

>
>> auto mode on (shift+tab to cycle) · ← for agents
```

Figure 5: Bedrock quotas retrieved live from the account (us-east-1)

Model	RPM	TPM	Notes
Sonnet 4.6	10	6M	default RPM is 10,000
Sonnet 4.5	10	5M	
Sonnet 4.5 (1M context)	1	1M	27M tokens/day cap
Haiku 4.5	10	5M	
Opus 4.6	5	3M	
Opus 4.5	5	2M	13.5M tokens/day (default 2.88B)
Claude 3.5 Haiku	10	2M	on-demand, non-adjustable

RPM is your binding constraint, not TPM. At 10 RPM with 6M TPM on Sonnet 4.6, you'd need 600K tokens per request to saturate the token budget – you'll hit request throttling long before that. Any concurrent workload will throttle almost immediately.

Models at zero quota (unusable until raised)

Claude Opus 5, Sonnet 5, Opus 4.7, Opus 4.8, Fable 5, Fable 5.1 – all at 0 TPM on both cross-region and global profiles. Same for GPT-5.4/5.5/5.6/6, Grok 4.6, and every bedrock-mantle endpoint quota (the OpenAI-compatible Responses/Chat Completions API is entirely unavailable).

Also heavily cut: most Llama, Mistral, Nova, Cohere, and Stability quotas sit at ~1% of default (e.g. Llama 3.1 70B at 4 RPM vs 400; Nova Canvas at 1 RPM vs 100).

What to do [Jump to bottom \(click\) ⓘ](#)

Update available! Run: brew upgrade claude-code@latest

>

>> auto mode on (shift+tab to cycle) · ← for agents

Figure 6: Per-model RPM/TPM limits and models currently at zero quota

Write Permissions Test

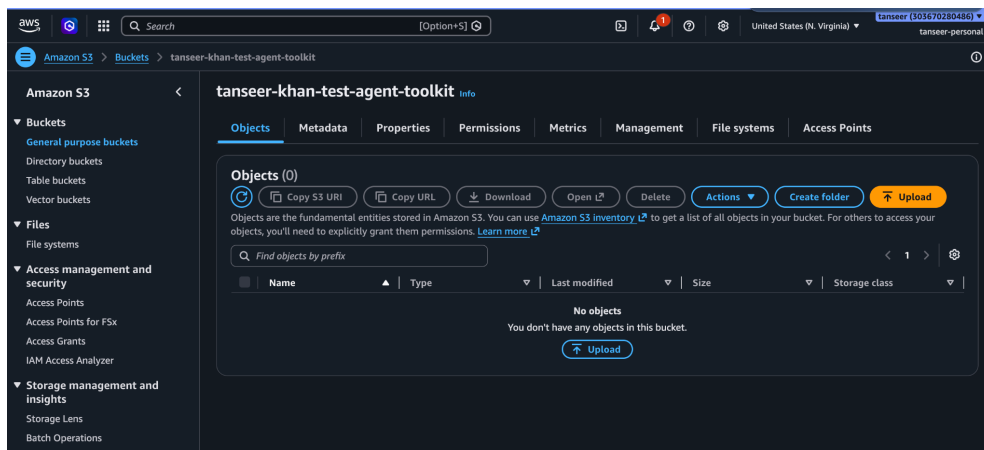
Finally, the agent was asked to make a change in the account to prove it has write access:

“Create a test S3 bucket called tanseer-khan-test-agent-toolkit”

The agent created the bucket in us-east-1 through aws-mcp and verified its default configuration:

- Block Public Access: all four settings enabled.
- Default encryption: SSE-S3 (AES256), with SSE-C blocked.
- Location constraint: null, which is how S3 represents us-east-1.

The bucket was then confirmed independently in the AWS Management Console (Amazon S3 › Buckets), where it appears as a new, empty bucket in United States (N. Virginia).



```
What to do

1. Request increases in the Service Quotas console for the specific model + profile you use – adjustable quotas are marked so (all the Claude 4.5/4.6/5 entries are). The 0-value ones are adjustable too, so they can be opened up.
2. Set maxTokens explicitly on every call. Leaving it unset reserves the model's full output maximum against your TPM budget, which at these limits will throttle you on the first request.
3. Verify billing is in good standing – across-the-board reductions like this often trace back to account verification rather than anything you configured.

Want me to check us-west-2 (or another region) for comparison, or pull the CloudWatch invocation metrics to see which quotas you're actually hitting?

* Worked for 2m 58s · done 11:14 AM

Create a test S3 bucket called tanseer-khan-test-agent-toolkit

• Creating it in us-east-1 (matching the region we've been working in).
  Called aws-mcp

• Created: tanseer-khan-test-agent-toolkit in us-east-1.

Verified defaults on the new bucket:
- Block Public Access – all four settings on (BlockPublicAcls, IgnorePublicAcls, BlockPublicPolicy, RestrictPublicBuckets)
- Default encryption – SSE-S3 (AES256), with SSE-C blocked
- Location constraint – null, which is how S3 represents us-east-1

Nothing else was configured – no versioning, lifecycle, logging, or tags. Since this is a test bucket, worth knowing: it has no lifecycle rule, so anything you put in it stays (and bills) until you delete it. Say the word if you want a lifecycle expiration, versioning, or a different region, and I'll add it or recreate.

* Churned for 20s · done 11:16 AM

Update available! Run: brew upgrade claude-code@latest

> Enable versioning on it

>> auto mode on (shift+tab to cycle) · ← for agents
```

Figures 7–8: The new bucket in the S3 console, and the agent’s creation and verification output

Summary of Results

Test	Result	Evidence
Installation and configuration	Passed: 23 skills installed and AWS MCP server configured	Figures 1–2
Basic connectivity	Passed: all 37 Regions returned through aws-mcp	Figure 3
Live data (not memory)	Passed: current EC2 families and account-specific Bedrock quotas	Figures 4–6
Write permissions	Passed: S3 bucket created and confirmed in the console	Figures 7–8

Conclusion

Claude Code is successfully connected to the AWS account through the AWS Agent Toolkit and the AWS MCP server. The agent can read live AWS documentation and account data, and it has the write permissions needed to create resources.

Next steps: request Bedrock quota increases before running real workloads, and delete the test bucket (or add a lifecycle rule) once testing is complete.